

Abdul Wahid
CTI Lab Report

Passive OSINT Reconnaissance — Google Search Operators & GHDB Category Analysis

Executive Summary

This lab demonstrates a systematic approach to discovering publicly exposed vulnerabilities using Google Dorking techniques and the Google Hacking Database (GHDB). The methodology follows standard penetration testing reconnaissance phases, utilizing legal OSINT tools to identify potential security weaknesses in web applications and infrastructure.

The lab is divided into two major sections. The first section covers all Google Search Operators — from basic to advanced — with live search results demonstrating each operator in action. The second section covers GHDB category navigation and how to identify, read, and execute real dorks from Exploit-DB against targets.



Section 1 — Basic Google Search Operators

Google Search Operators are special commands that filter, narrow, or expand search results with precision. In GHDB context, they are used to find exposed, sensitive, or misconfigured information that Google has indexed on the internet.

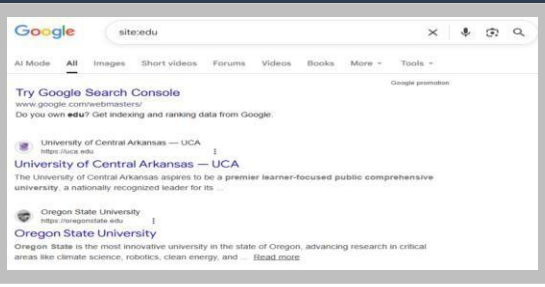
Operator	What It Does
site:	Restricts results to a specific domain or website
inurl:	Searches for a keyword inside the URL of a page
intitle:	Searches for a keyword inside the HTML <title> tag
intext:	Searches for a keyword inside the body/content of a webpage
filetype:/ ext:	Filters results to show only specific file types
cache:	Shows Google's cached (saved) version of a page
- (minus)	Excludes a keyword from results
" " (quotes)	Searches for an exact phrase
* (wildcard)	Acts as a placeholder for any word
OR	Returns results matching either of two terms

site: — Domain Restriction

Restricts all search results to a specific domain or website. This is the most fundamental recon operator — every targeted dorking session starts here.

```
site:example.com      → All indexed pages of a domain
site:gov              → All .gov domains
site:edu              → All educational institutions
```

Query: site:edu

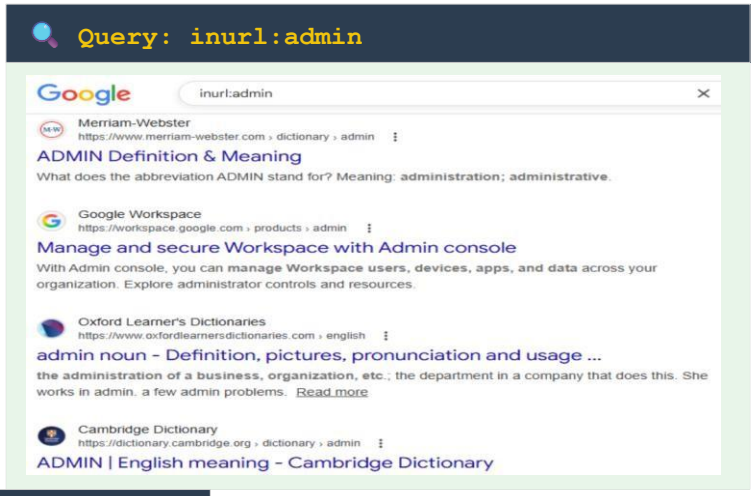


inurl: — URL Keyword Search

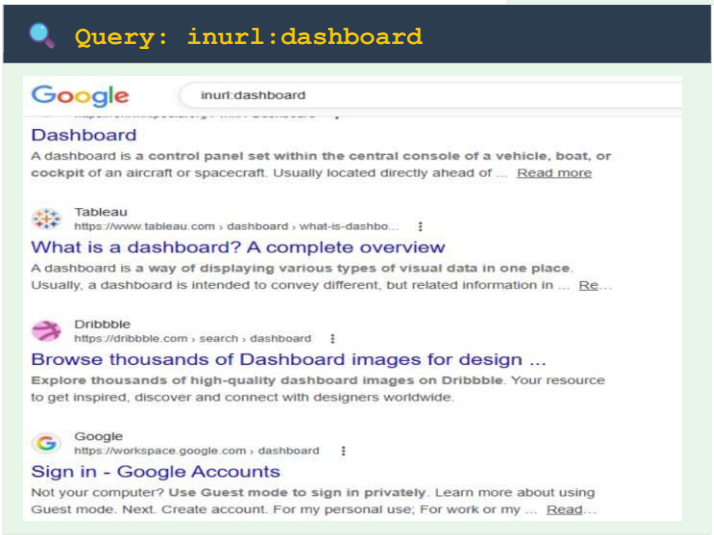
Searches for a specific keyword inside the URL path of a webpage. Extremely useful for finding admin panels, login pages, and sensitive endpoints that are publicly accessible.

```
inurl:admin          → Pages with 'admin' in the URL
inurl:login          → Login pages across the web
inurl:dashboard      → Dashboard interfaces
```

Query: inurl:admin



Query: inurl:dashboard

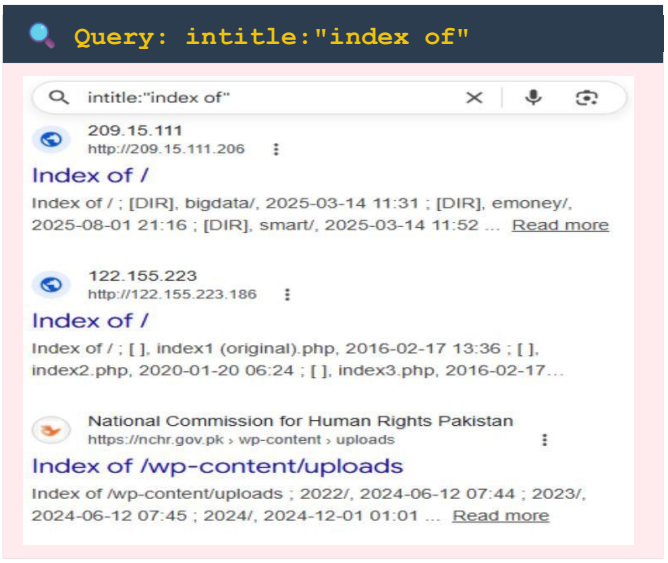



```
Use case: Find admin panels, login pages, dashboards exposed on the web.
```

intitle: — Page Title Search

Searches for keywords inside the HTML <title> tag of a webpage. One of the most powerful operators for finding open directory listings and exposed control panels, since page titles often reveal the purpose of the page.

```
intitle:"index of"    → Open directory listings
intitle:"admin panel" → Admin control panels
intitle:"login page"  → Login interfaces
```

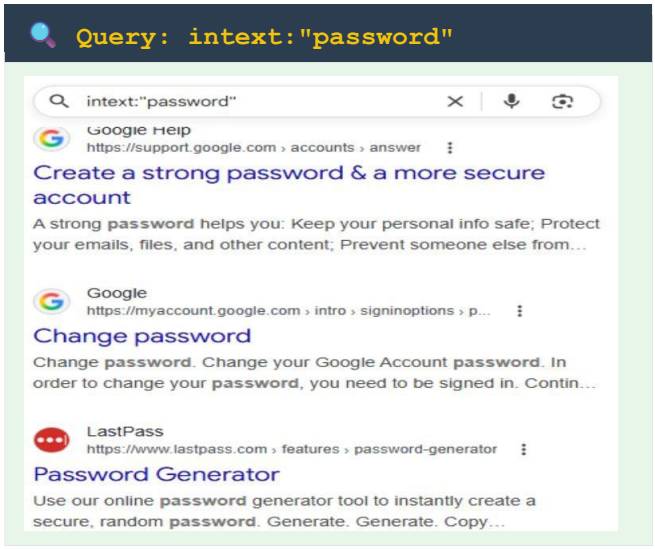


⚠ High Value: The intitle:"index of" dork is one of the most famous in GHDB. It reveals servers with directory listing enabled — potentially exposing all files in that directory to anyone.

intext: — Page Body Content Search

Searches for keywords inside the visible body content of a webpage. Useful for finding pages that contain sensitive strings like passwords, error messages, or confidential data that has been accidentally exposed.

```
intext:"password"        → Pages containing the word password
intext:"confidential"    → Pages with confidential content
intext:"not for public" → Pages with restricted notices
```



Use case: Find pages containing sensitive keywords in body content. Power comes from combining with filetype: and site: operators.

filetype: / ext: — File Type Filter

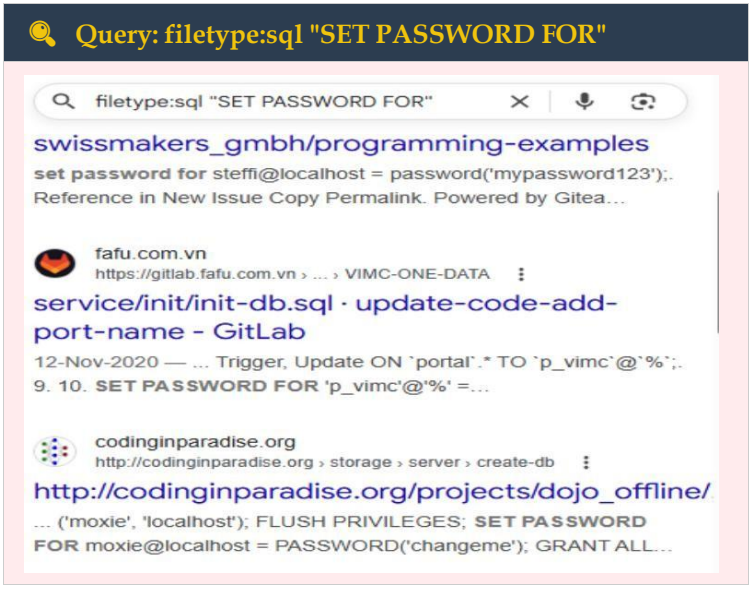
Filters search results to show only specific file types. This is one of the highest-value operators in GHDB because many sensitive files are accidentally left publicly accessible on web servers.

```
filetype:pdf     → PDF documents
filetype:sql     → SQL database files
filetype:env     → Environment configuration files
ext:log          → Log files
ext:bak          → Backup files
```

Query: NCCIA filetype:pdf



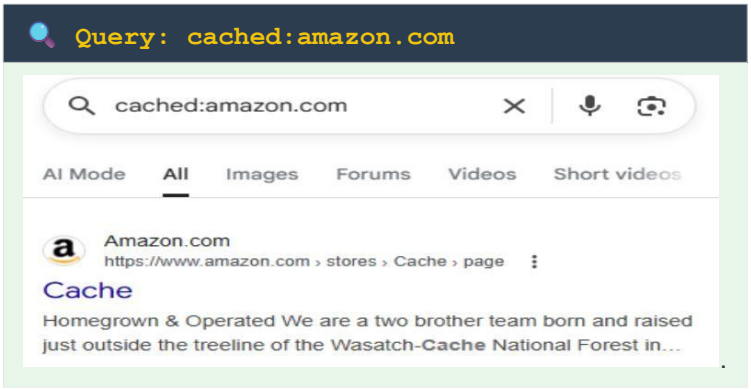
Query: filetype:sql "SET PASSWORD FOR"



cache: — Google's Cached Version

Displays Google's cached (saved) version of a webpage. Useful in recon because it allows you to view page content even if the target has taken it down since Google last crawled it.

```
cache:example.com      → View Google's saved copy of a page
cache:amazon.com       → Amazon's cached version
```

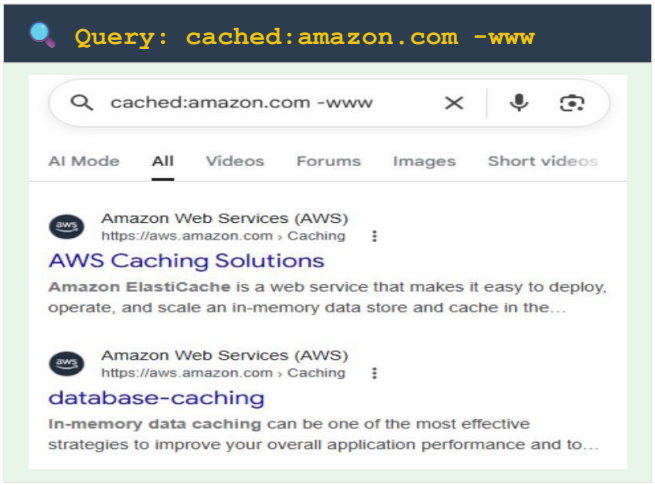


Use case: View a page even after it has been taken down — critical for recon as targets may remove exposed files after discovery.

- (Minus) — Keyword Exclusion

Excludes specific keywords from search results to eliminate noise and irrelevant results. Essential for precision dorking when generic queries return too many unrelated results.

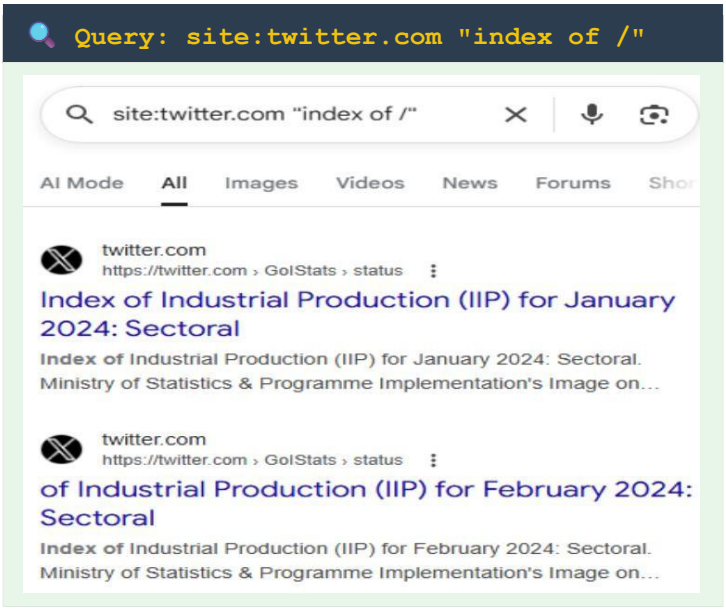
```
site:example.com -www          → Exclude www subdomain
inurl:admin -site:youtube.com  → Admin pages but not YouTube
```



" " (Exact Quotes) — Exact Phrase Match

Searches for an exact phrase, word-for-word as typed inside the quotes. One of the most powerful operators in GHDB because it allows finding specific error messages, file contents, and exposed strings precisely.

```
"index of /"              → Exact directory listing string
"admin password"          → Exact phrase in page content
"error in your SQL syntax" → Exact SQL error message
```

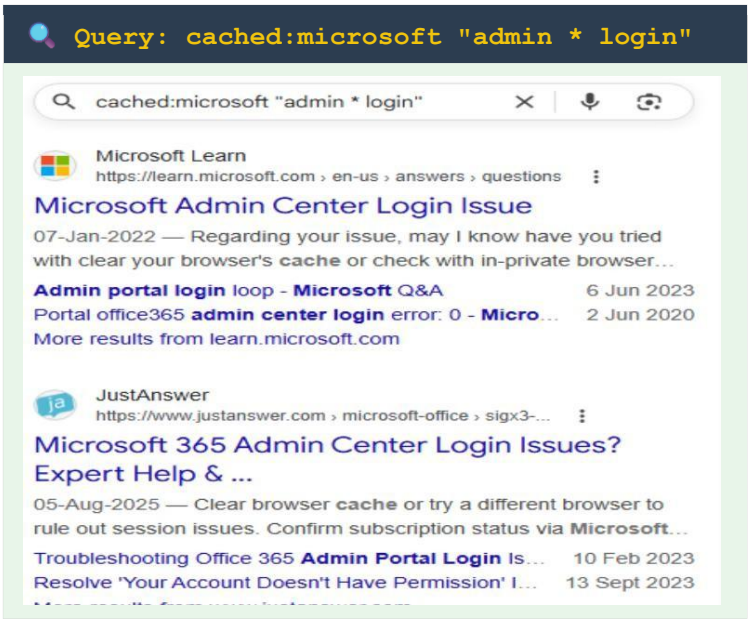


Use case: Find exact strings in page content — particularly powerful for finding specific error messages, default page text, and exposed file directory headers.

Y. * (Wildcard) — Placeholder

Acts as a placeholder for any word or phrase. Allows finding variations of a search pattern without knowing the exact middle terms. Useful for casting a wider net.

```
"admin * login"      → admin [anything] login
"error * database"   → error [anything] database
```

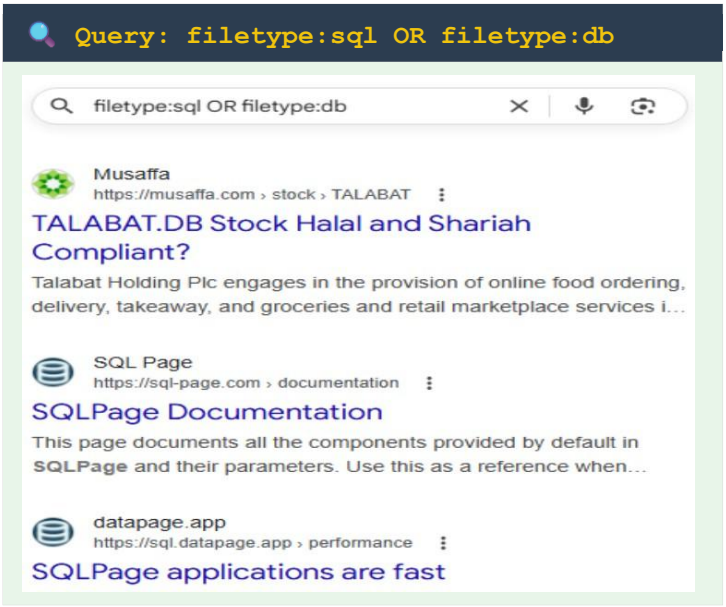


Use case: Cast a wider net when not sure of exact phrasing. Particularly useful in GHDB for finding error message variations.

10. OR — Boolean OR Operator

Returns results matching either of two or more terms. Allows combining multiple search possibilities into a single query — increasing coverage without running multiple separate searches.

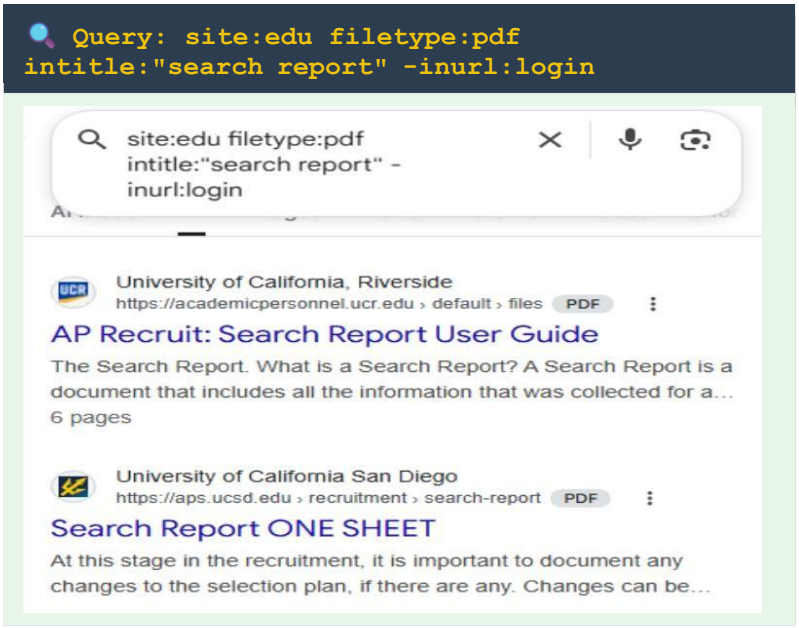
```
inurl:admin OR inurl:administrator  → Admin OR administrator in URL
filetype:sql OR filetype:db         → SQL or DB files
```



Complete Combined Query — Power Dorking

The true power of Google operators emerges when combining multiple operators into a single precise query. This is the foundation of GHDB dorking.

```
site:edu filetype:pdf intitle:"search report" -inurl:login
```



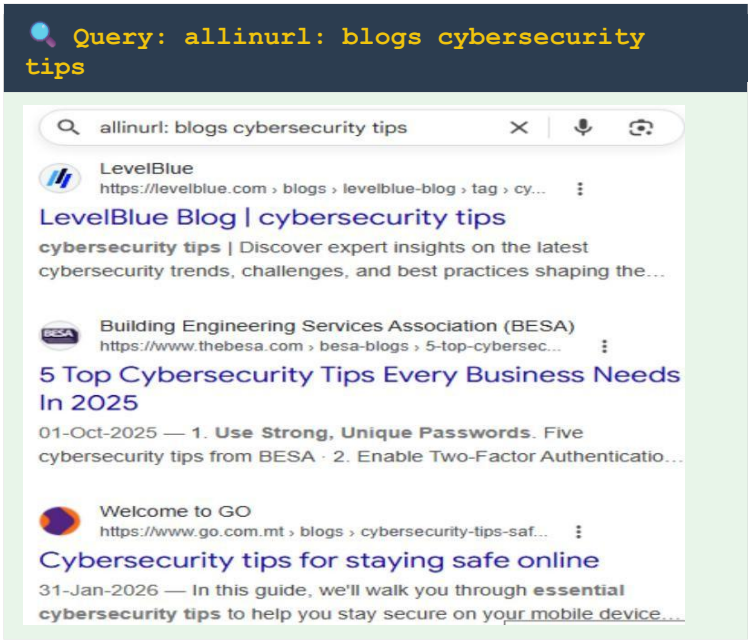
Section 2 — Intermediate Google Search Operators

Intermediate operators are more specific versions of basic operators. They require all keywords to match specific page elements and are used for highly targeted searches.

allinurl: — All Keywords in URL

All specified keywords must appear in the URL. Unlike inurl: which checks for one keyword, allinurl: enforces all keywords to be present in the URL simultaneously.

```
allinurl: blogs cybersecurity tips
```

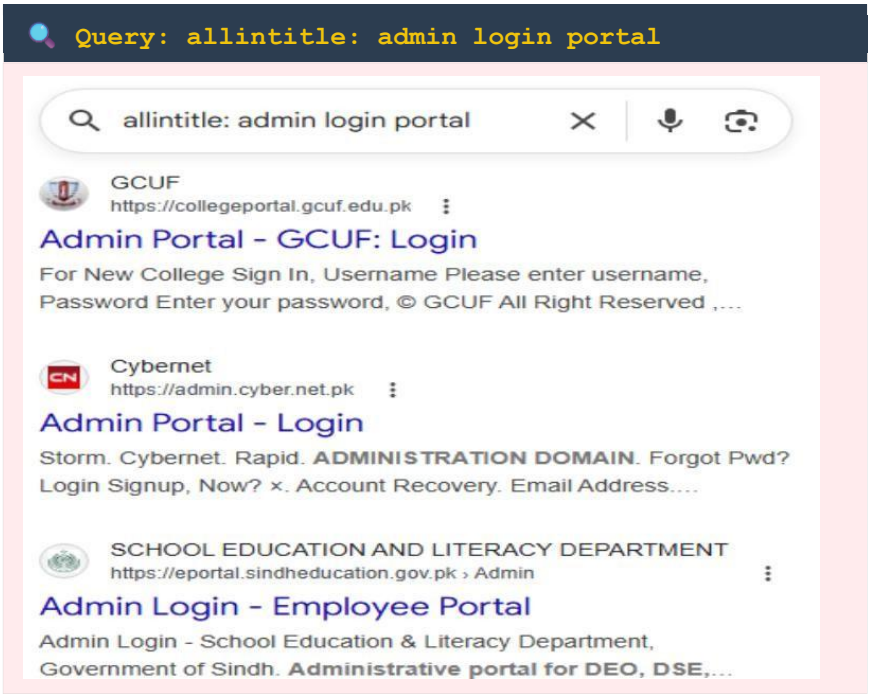



```
Use case: Finding blog URLs specifically about cybersecurity tips.
The allinurl: operator cannot be mixed with other operators
effectively — use separate inurl: operators for combining.
```

allintitle: — All Keywords in Page Title

All specified keywords must appear in the HTML page title. More precise than intitle: for multi-word title searches. Excellent for finding specific admin interfaces.

```
allintitle: admin login portal
```

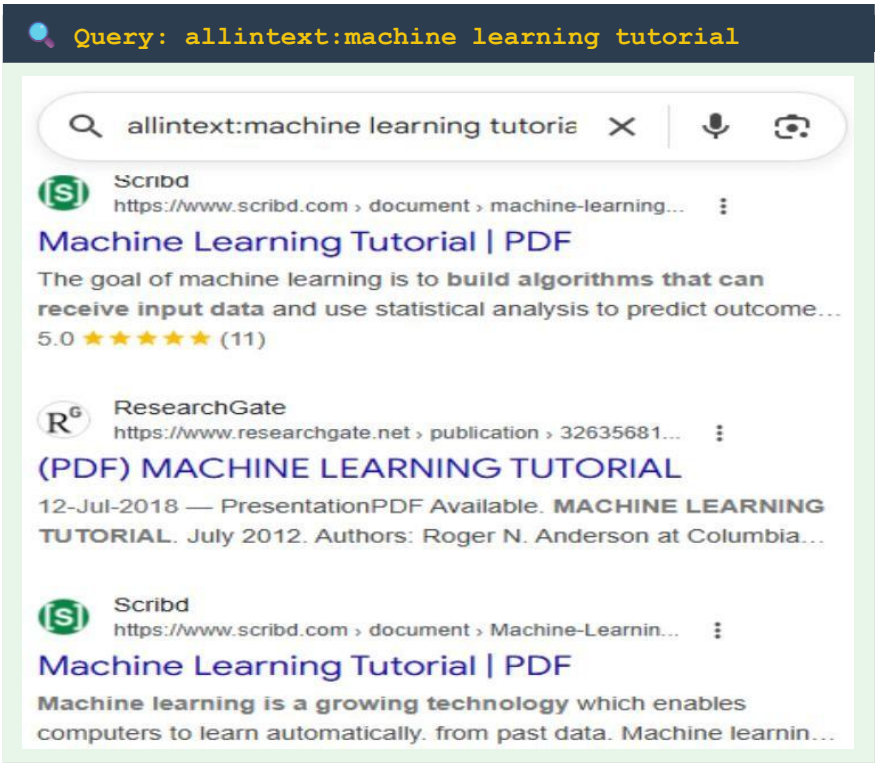


Use case: Finding pages whose title clearly includes 'admin login portal'. Results are highly targeted — every result will have all three words in the title.

allintext: — All Keywords in Body Content

All specified keywords must appear in the body content of the webpage. More strict than intext: — ensures complete context matching rather than individual keyword presence.

```
allintext:machine learning tutorial
```

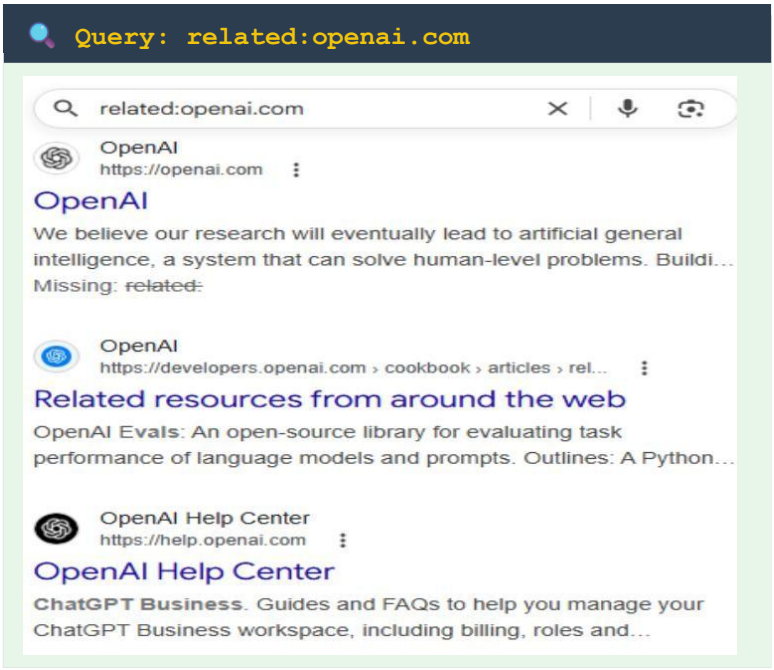


Use case: Finding detailed beginner-level tutorials, or in security context — finding pages where multiple sensitive terms co-exist in the page body.

related: — Similar Websites

Shows websites that Google considers similar or related to a given domain. Useful in recon for finding subsidiary companies, partners, or competitor infrastructure associated with a target organization.

```
related:openai.com
```

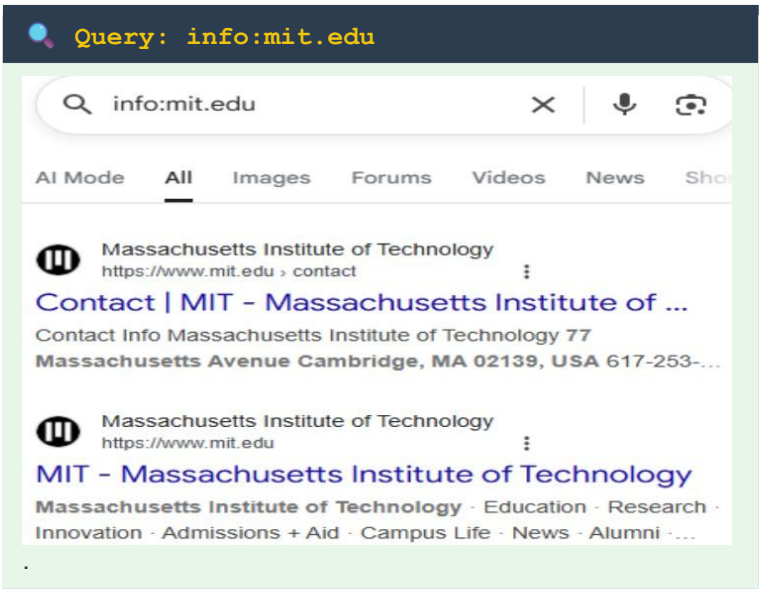


Use case: Finding competitors or similar AI platforms. In pentesting: discovering sister sites, partner domains, or associated infrastructure of a target.

info: — Google's Indexed Information

Displays Google's stored information about a specific URL — including its cached version, related pages, and indexed details. Useful as a quick reconnaissance starting point for any target.

```
info:mit.edu
```

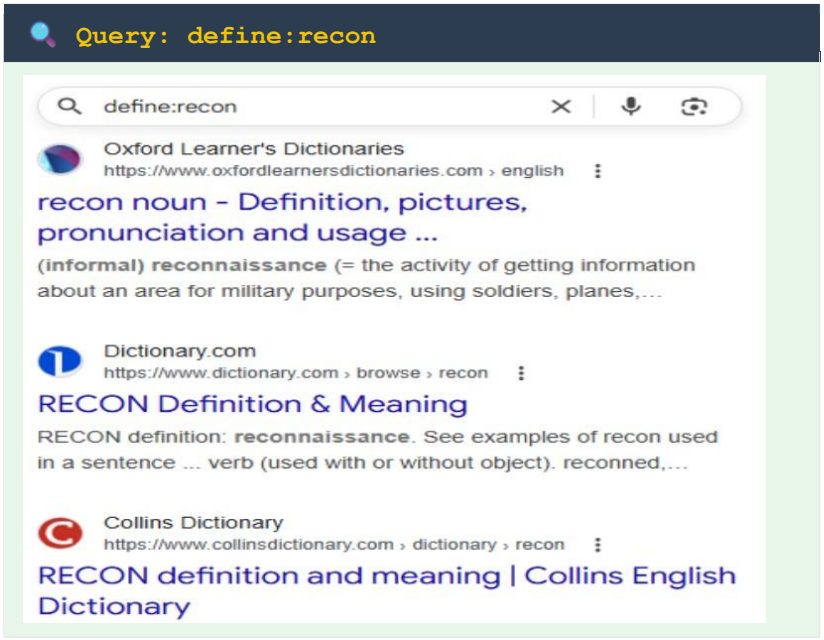


Use case: Checking indexed version, related pages, and cache access of a target domain. Quick recon snapshot.

define: — Word Definition

Returns the dictionary definition of a word or term directly in search results. While less used in offensive GHDB, it is useful for quickly looking up security terminology during research.

```
define:recon
```



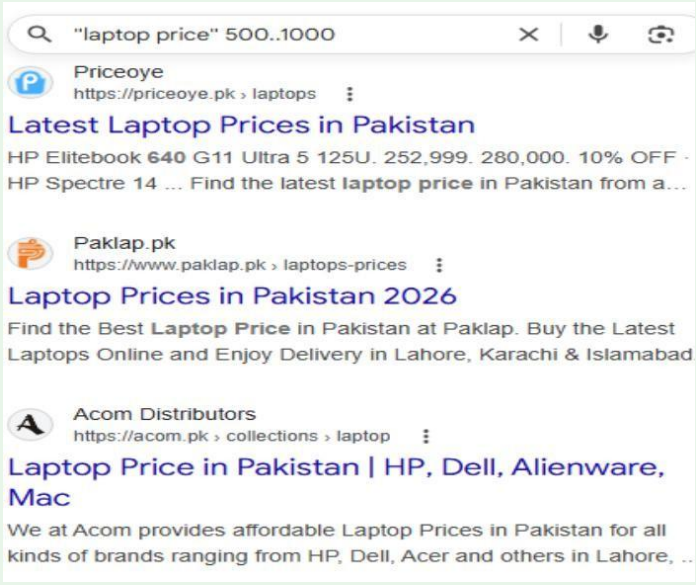

```
Use case: Quick terminology lookup during research without
navigating away from the browser.
```

numrange: / .. — Number Range

Filters search results to include only content containing numbers within a specified range. The double-dot (..) syntax is the most commonly used form. Useful for filtering price ranges, port numbers, version numbers, or year ranges.

```
"laptop price" 500..1000     → Laptops priced between 500 and 1000
port 8000..9000               → References to ports in this range
```

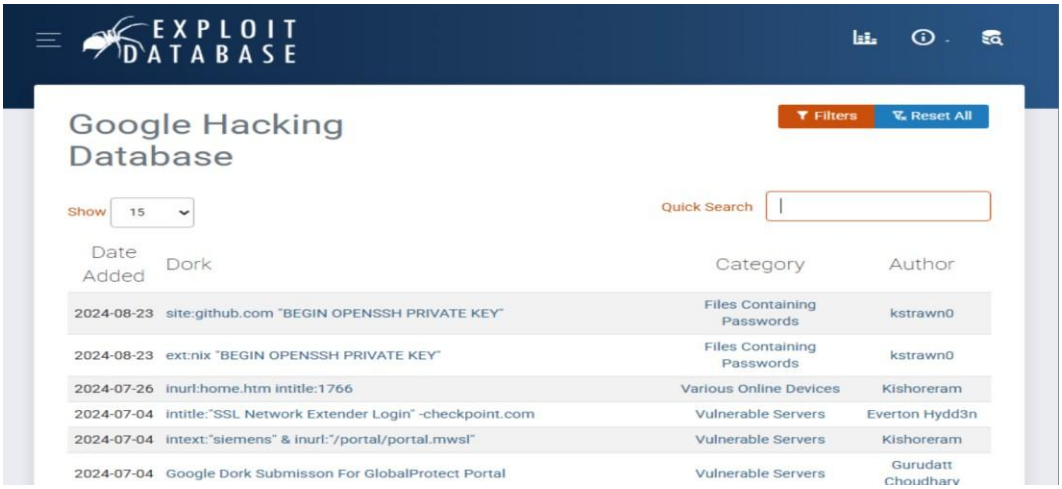
🔍 Query: "laptop price" 500..1000



Use case: Filtering numeric data within a specific range. In security: filtering results by year, version number, or port range.

Section 3 — Google Hacking Database (GHDB) Navigation

The Google Hacking Database (GHDB) is a centralized repository hosted on Exploit-DB (exploit-db.com/google-hacking-database) containing thousands of community-contributed Google dorks organized by vulnerability category. Understanding how to navigate and use GHDB is essential for professional recon.



Feature	Detail
Platform	Exploit-DB — exploit-db.com/google-hacking-database
Total Dorks	7,944+ active dorks (as of 2026)
Date Range	Publications from 2003 to present
Contribution	Community-contributed security queries
Organization	Categorized by vulnerability type
Search Features	Quick Search, Category Filter, Author Filter

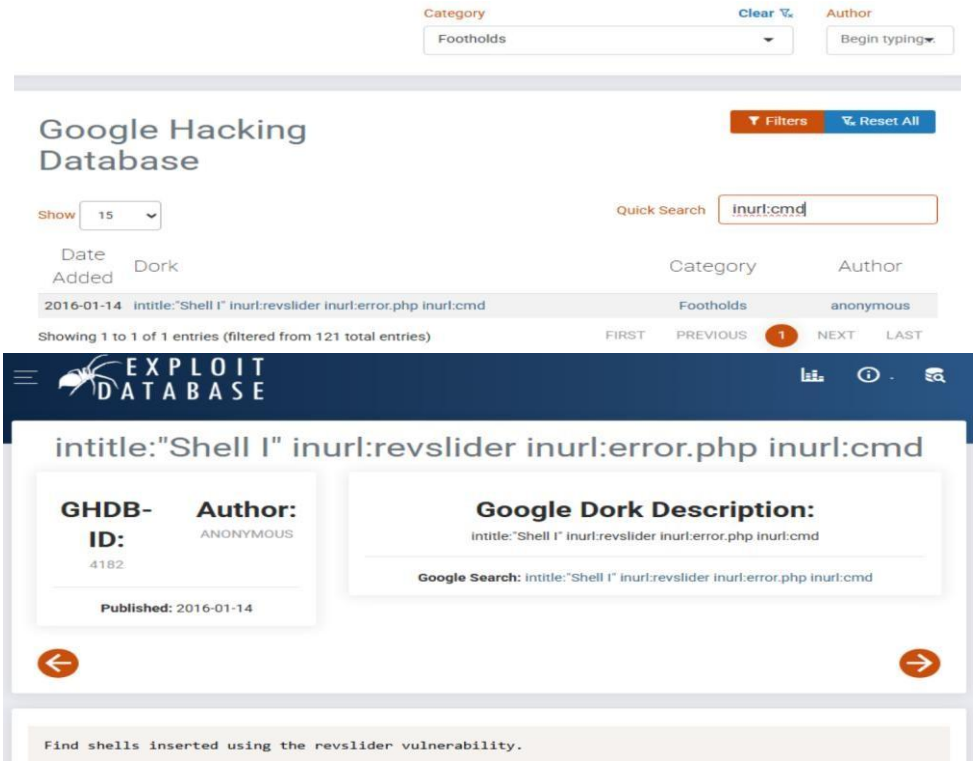
GHDB Categories — Deep Dive

GHDB organizes all dorks into specific vulnerability categories. Each category targets a different type of security weakness. The following section covers each category with its GHDB interface view and live Google results.

1. Footholds

Meaning	The first step in any engagement — finding an entry point into the target system. Think of it as finding an open window or unlocked door in a building.
What You Find	Exposed admin panels, web shells, RDP/SSH web interfaces, publicly accessible control interfaces
GHDB Result	intitle:"Shell I" inurl:revslider inurl:error.php inurl:cmd → GHDB-ID: 4182, Author: Anonymous, Published: 2016-01-14
Google Output	Found beneri.se with URL: beneri.se/uploads/admin/.../cmd/error.php/revslider — an actual web shell accessible via the Revslider vulnerability. The directory listing showed cmd and error.php files — confirming a live foothold.

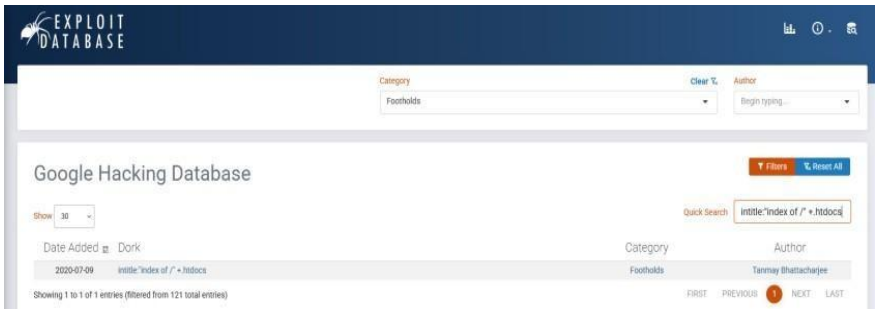
Test-1

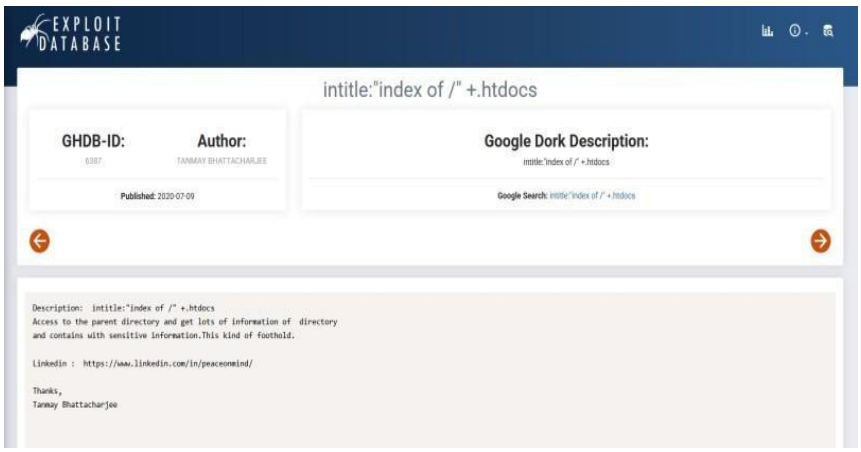



intitle:"Shell I" inurl:revslider inurl:error.php inurl:cmd
AI Mode
All
Videos
Short videos
Forums
Images
Web
More
Tools
beneri.se
https://beneri.se › cmd › revslider · Translate this page
Shell I - Beneri
Shell I revslider error.php cmd.


beneri.se/uploads/admin/29f51109899979326836d79fe2be28f7/cmd/error.php/revslider
Shell I revslider error.php cmd

Test-2

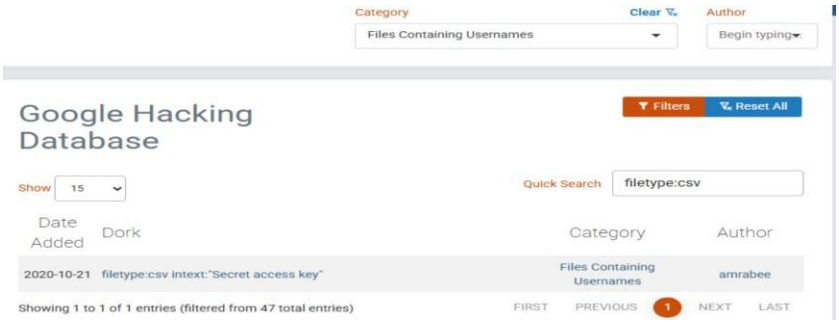

EXPLOIT DATABASE
Category
Footholds
Author
Google Hacking Database
Filters
Reset All
Show 30
Quick Search
intitle:"index of /" +.htdocs
Date Added
Dork
Category
Author
2020-07-09
intitle:"index of /" +.htdocs
Footholds
Tanmay Bhattacharjee
Showing 1 to 1 of 1 entries (filtered from 121 total entries)
FIRST
PREVIOUS
1
NEXT
LAST

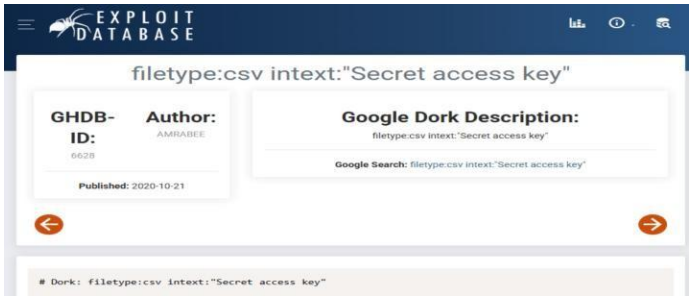

EXPLOIT DATABASE
intitle:"index of /" +.htdocs
GHDB-ID:
Author:
TANMAY BHATTACHARJEE
Google Dork Description:
intitle:"index of /" +.htdocs
Published: 2020-07-09
Google Search: intitle:"index of /" +.htdocs
Description: intitle:"index of /" +.htdocs
Access to the parent directory and get lots of information of directory
and contains with sensitive information.This kind of foothold.
Linkedin : https://www.linkedin.com/in/peaceonmind/
Thanks,
Tanmay Bhattacharjee

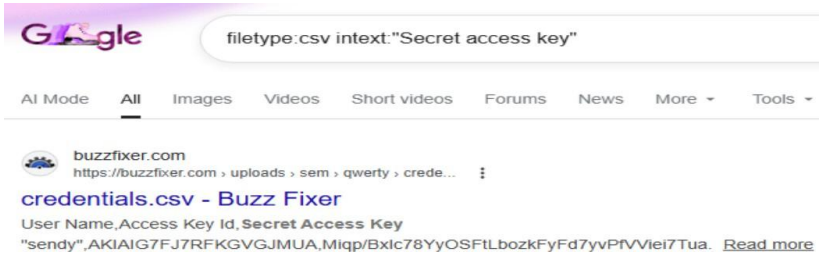

intitle:"index of /" .htdocs
AI Mode
All
Images
Shopping
Videos
Short videos
Forums
More
Tools
NOAA (.gov)
https://ftp.cpc.ncep.noaa.gov › htdocs
Index of /htdocs
Index of /htdocs. Name · Last modified · Size · Parent Directory, -. archives.wkthurs/, 05-May-2008
19:09, -. cai/, 20-Feb-2007 21:46, -. data/, 11-Aug-2015 15: ...

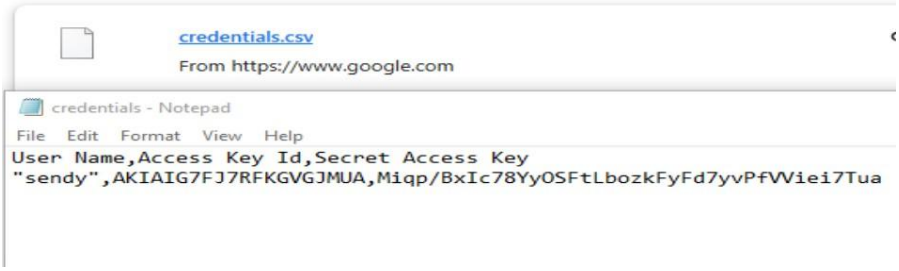
2. Files Containing Usernames

Meaning	Finding files where usernames are publicly exposed — when developers accidentally leave files containing user lists on public-facing servers.
What You Find	/etc/passwd files, User list files, CMS user exports, Database dumps containing usernames
GHDB Result	filetype:csv intext:"Secret access key" → GHDB-ID: 6628, Author: amrabee, Published: 2020-10-21
Google Output	CRITICAL — Google returned credentials.csv from buzzfixer.com containing actual AWS credentials: 'User Name, Access Key Id, Secret Access Key' with real keys like AKIAIG7FJ7RFKGVGJMUA visible in plaintext. File was opened in Notepad showing live AWS Secret Access Keys.



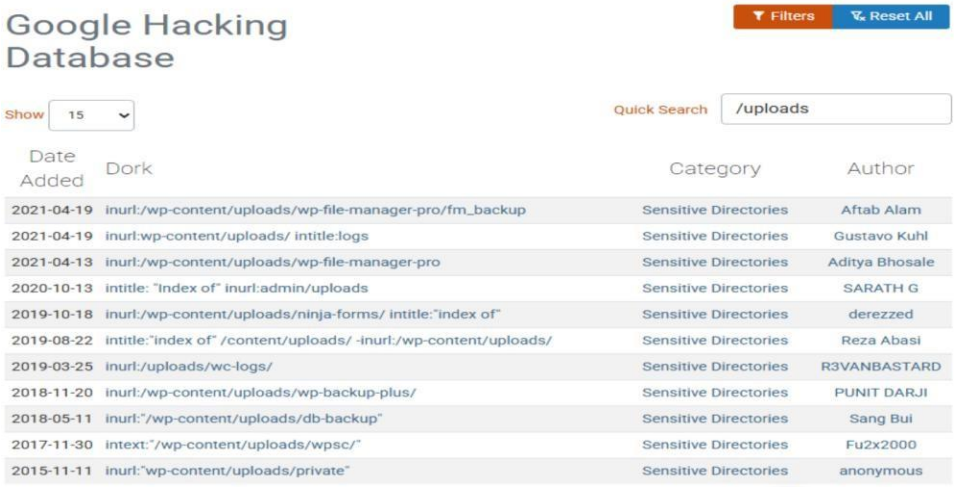






3. Sensitive Directories

Meaning	Finding folders that were never supposed to be publicly accessible — exposed directories that Google has indexed, allowing anyone to browse their contents.
What You Find	Backup folders (/backup), Upload directories (/uploads), Admin directories, Configuration folders, Private document folders
GHDB Result	inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup → GHDB-ID: 6878, Author: Aftab Alam, Published: 2021-04-19
Google Output	Google returned arctic-sdi.org showing an open directory listing 'Index of /wp-content/uploads/wp-file-manager-pro' with a visible fm_backup/ folder dated 2019-06-07. Anyone can browse and download all backup files from this exposed directory.

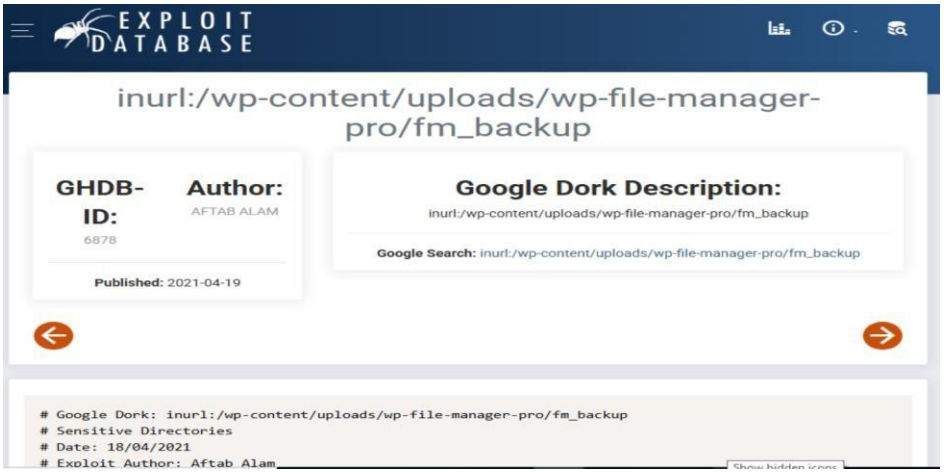


Google Hacking Database

Filters | Reset All

Show 15 | Quick Search: /uploads

Date Added	Dork	Category	Author
2021-04-19	inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup	Sensitive Directories	Aftab Alam
2021-04-19	inurl:wp-content/uploads/ intitle:logs	Sensitive Directories	Gustavo Kuhl
2021-04-13	inurl:/wp-content/uploads/wp-file-manager-pro	Sensitive Directories	Aditya Bhosale
2020-10-13	intitle: "Index of" inurl:admin/uploads	Sensitive Directories	SARATH G
2019-10-18	inurl:/wp-content/uploads/ninja-forms/ intitle:"index of"	Sensitive Directories	derezzed
2019-08-22	intitle:"index of" /content/uploads/ -inurl:/wp-content/uploads/	Sensitive Directories	Reza Abasi
2019-03-25	inurl:/uploads/wc-logs/	Sensitive Directories	R3VANBASTARD
2018-11-20	inurl:/wp-content/uploads/wp-backup-plus/	Sensitive Directories	PUNIT DARJI
2018-05-11	inurl:"/wp-content/uploads/db-backup"	Sensitive Directories	Sang Bui
2017-11-30	intext:"/wp-content/uploads/wpsc/"	Sensitive Directories	Fu2x2000
2015-11-11	inurl:"wp-content/uploads/private"	Sensitive Directories	anonymous



EXPLOIT DATABASE

inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup

GHDB-ID: 6878

Author: AFTAB ALAM

Published: 2021-04-19

Google Dork Description:

inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup

Google Search: inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup

```
# Google Dork: inurl:/wp-content/uploads/wp-file-manager-pro/fm_backup
# Sensitive Directories
# Date: 18/04/2021
# Exploit Author: Aftab Alam
```



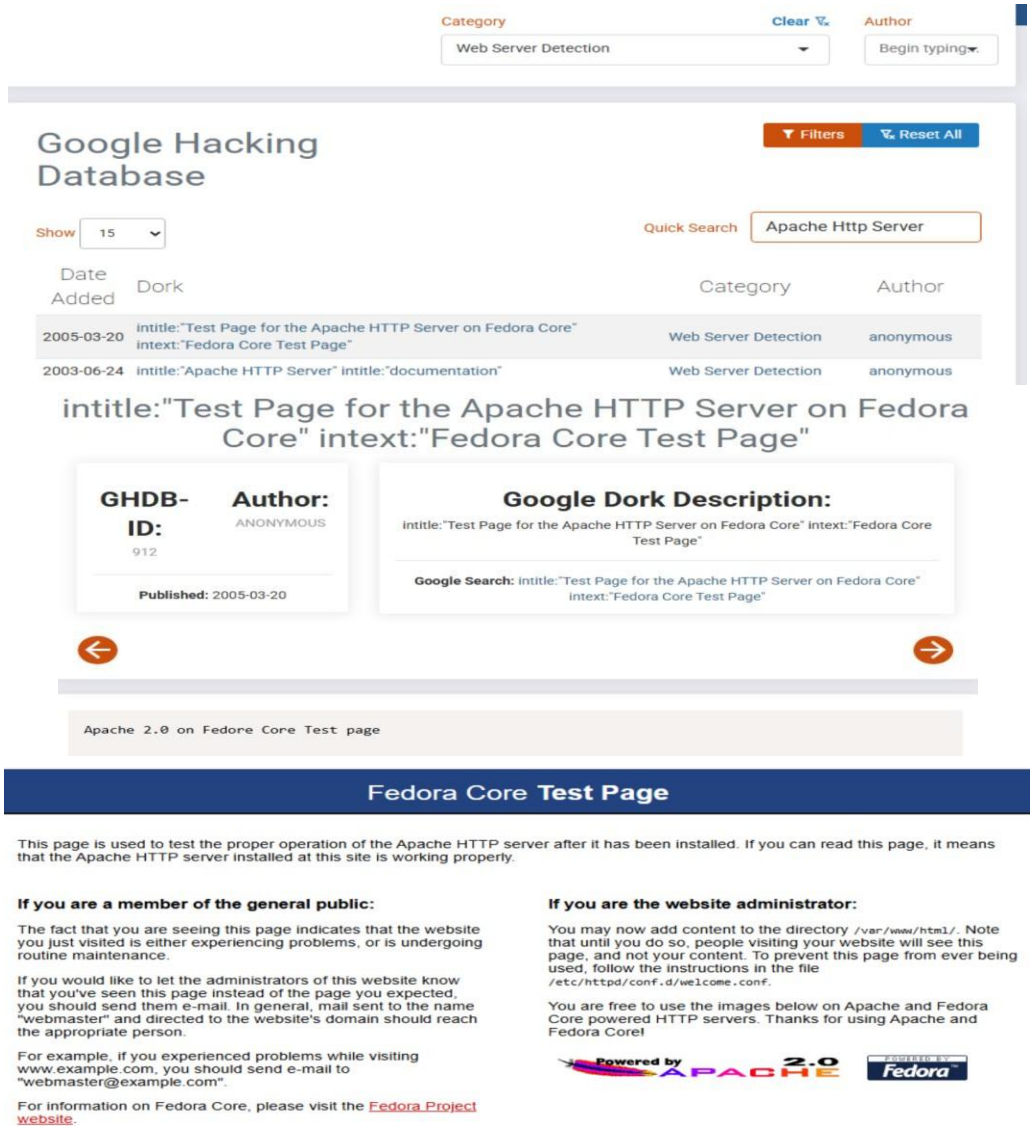
Index of /wp-content/uploads/wp-file-manager-pro

Name	Last modified	Size	Description
Parent Directory		-	
fm_backup/	2019-06-07 14:15	-	

Apache/2.4.66 (Debian) Server at arctic-sdi.org Port 443

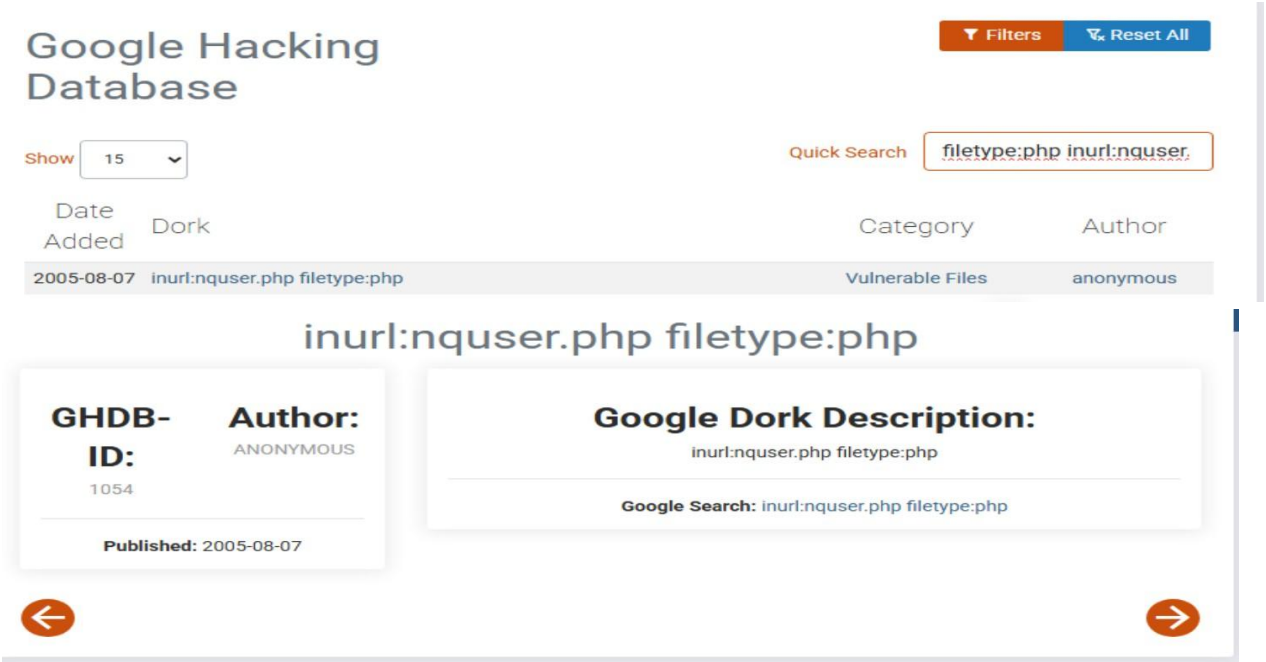
4. Web Server Detection

Meaning	Identifying what web server technology a target is running — version, configuration, and default pages that reveal server infrastructure details.
What You Find	Apache version info, Nginx configuration, IIS default pages, Tomcat interfaces, Fedora/Ubuntu default test pages
GHDB Result	intitle:"Test Page for the Apache HTTP Server on Fedora Core" intext:"Fedora Core Test Page" → GHDB-ID: 912, Author: Anonymous, Published: 2005-03-20
Google Output	Google returned the Fedora Core Test Page — a default Apache installation page showing 'Apache 2.0 on Fedora Core Test page'. This reveals: (1) Apache 2.0 server version, (2) Fedora Core OS, (3) The admin never customized the server after installation — indicating a potentially poorly maintained server.

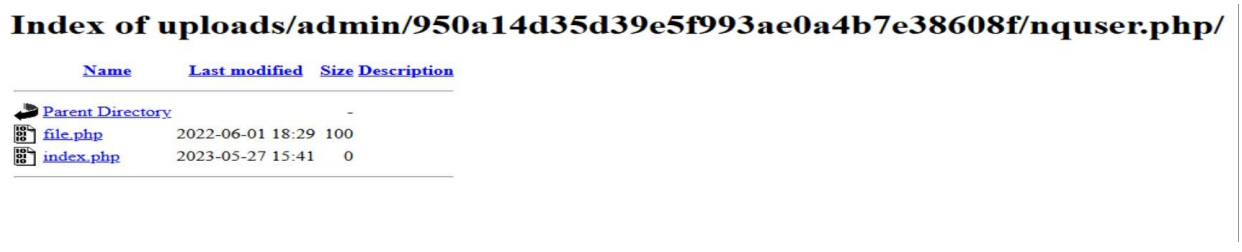


5. ⚠ Vulnerable Files

Meaning	Directly finding files on web servers that contain known vulnerabilities or expose sensitive data that should never be publicly accessible.
What You Find	PHP files with remote code execution, configuration files with credentials, files vulnerable to path traversal, exposed API endpoints
GHDB Result	inurl:nquser.php filetype:php → GHDB-ID: 1054, Author: Anonymous, Published: 2005-08-07
Google Output	CRITICAL — Google returned an exposed Netquery 3.1 installation with remote command execution capability. The directory listing showed: Index of uploads/admin/950a14d35d39e5f993ae0a4b7e38608f/nquser.php/ with file.php and index.php files. The dork description confirms: 'user can view clear text log file, execute PING commands, and path traversal to view /etc/passwd'

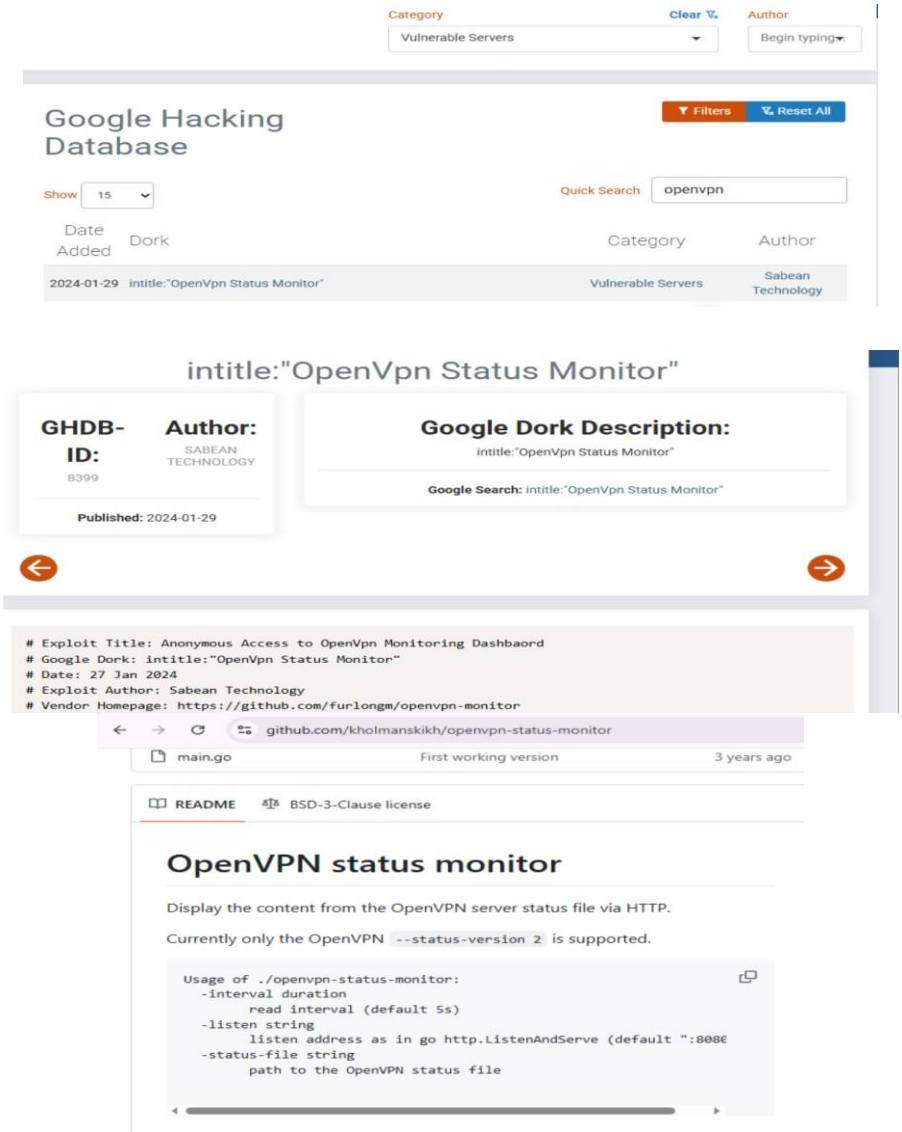



```
Netquery 3.1 remote commands execution, cross site scripting, information disclosure poc exploit
software: author site: http://www.virtech.org/tools/ a user can execute command on target system by
PING panel, if enabled like often happens, using pipe char on input text "Ping IP Address or Host
Name", example: | cat /etc/passwd then you will see plain text password file | pwd to see current path
| rm [pwd_output]/logs/nq_log.txt to delete log file... disclosure of user activity: if enabled, a
user can view clear text log file through url: http://[target]/[path]/logs/nq_log.txt xss:
http://[target]/[path]/submit.php?portnum="/>alert(document.cookie)
```



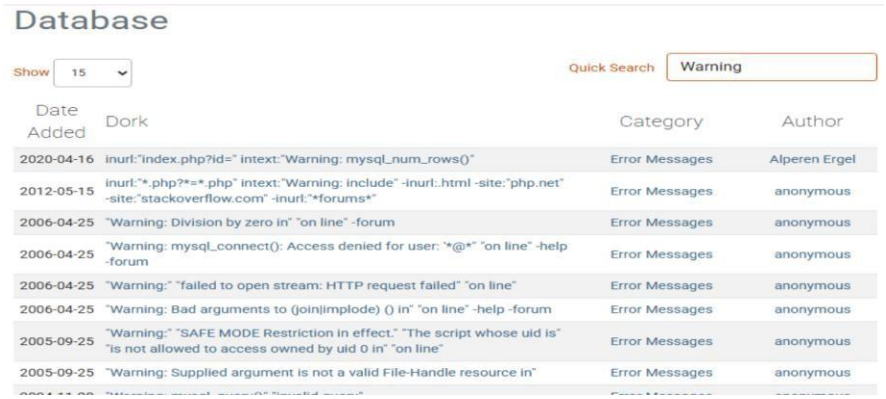
6. 💀 Vulnerable Servers

Meaning	Finding entire servers that have known CVEs, default credentials, or critical misconfigurations — not just individual files, but the entire server infrastructure.
What You Find	OpenVPN monitoring dashboards, default credential interfaces, unpatched CMS installations, exposed management consoles
GHDB Result	intitle:"OpenVpn Status Monitor" → GHDB-ID: 8399, Author: Sabean Technology, Published: 2024-01-29
Google Output	Google returned an OpenVPN Status Monitor dashboard with anonymous public access. The GHDB description: 'Anonymous Access to OpenVpn Monitoring Dashboard'. The GitHub link (github.com/furlongm/openvpn-monitor) confirms this is a real tool — and any publicly exposed instance allows anyone to view all connected VPN clients, their real IPs, and bandwidth usage.

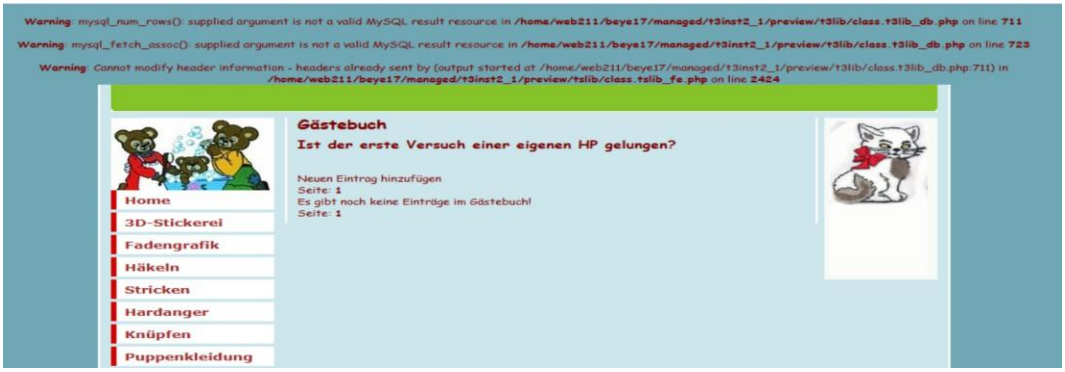


7. ❌ Error Messages — Information Leakage via Errors

Meaning	Application error messages that expose database names, file paths, server versions, and internal code structure
What You Find	MySQL errors, PHP warnings, stack traces, database connection strings, internal file paths, server configuration details
GHDB Dork	inurl:"index.php?id=" intext:"Warning: mysql_num_rows()" → GHDB-ID: 5881, Author: Alperen Ergel, Published: 2020-04-16
Google Result	🟢 CRITICAL — Google found a page at /home/web211/beye17/managed/t3inst2_1/preview/t3lib/ exposing multiple MySQL warnings: 'mysql_num_rows() supplied argument is not a valid MySQL result resource on line 711', 'mysql_fetch_assoc() on line 723'. The full server file path /home/web211/beye17/managed/ is exposed — revealing the exact directory structure of the web server.







Section 4 — Summary & Key Takeaways

— Operators

Operator	Primary Security Use
site:	Domain scope restriction — foundation of all targeted recon
inurl:	URL path keyword search — finds exposed endpoints
intitle:	Page title search — finds directory listings and panels
intext:	Body content search — finds sensitive strings
filetype:/ext:	File type filter — finds .env, .sql, .log, .bak files
cache:	Cached page access — view taken-down content
- (minus)	Noise exclusion — precision filtering
" "	Exact phrase match — finds specific error strings
*	Wildcard placeholder — broad pattern matching
OR	Boolean OR — multiple possibilities in one query
allinurl:	All keywords in URL — precision URL targeting
allintitle:	All keywords in title — specific panel finding
allintext:	All keywords in body — context-complete search
related:	Similar sites — infrastructure mapping
info:	Google info about URL — quick recon snapshot
define:	Definition lookup — research tool
numrange/..	Number range filtering — version/port/year filtering

— GHDB Categories Covered

Category	What It Exposes
Footholds	Entry points, web shells, control interfaces
Files Containing Usernames	passwd files, CSV exports, user dumps
Sensitive Directories	wp-uploads, backup folders, private dirs
Web Server Detection	Apache/IIS/Nginx version and default pages
Vulnerable Files	RCE-vulnerable PHP, config files, API endpoints
Vulnerable Servers	OpenVPN dashboards, default credential interfaces
Error Messages	MySQL warnings, PHP errors, stack traces